

Direct Answer and Evidence: GREEN-028

Gowers Box Norms over Finite Fields

Direct Answer

Verdict: UNRESOLVED. The requested correlation statement remains open; no proof and no counterexample are known.

Proof status: not obtained.

Counterexample status: cubic-mixed-twist route refuted: the twist collapses the spectral hypothesis as $(2p - 1)/p^2 \rightarrow 0$ (proved); no counterexample known; numerical search $p = 5, 7$ not exhaustive.

Exact missing implication: The Fourier-mass hypothesis on B_f is not known to imply algebraic correlation with the infinite-dimensional target class; no known implication from the spectral condition to correlation exists; the collapse law shows the hypothesis suppresses nondegenerate mixed-bilinear components of diagonal increments; the naive converse classification is disproven (R8), so the open residue for reading (i) comprises the uniform twisted trilinear inequality (O1) and the corrected converse classification with cycle-twist components (O2) — a literature assessment, not a deduction proved here

Scope: General theorem for all odd primes p and dimensions n ; partial results hold where stated.

All numerical residuals quoted here were independently confirmed by an external deterministic verification suite, which was independently revalidated at build time during the preparation of this document.

Unconditional results

R1. Spectral reformulation (Proposition A). With $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1)\overline{f(x_2, y_1)}\overline{f(x_1, y_2)}f(x_2, y_2)$ on G^4 and $\Xi_0 = \{(\xi_1, \xi_2, \xi_3, \xi_4) \in$

$\widehat{G}^4 : \xi_1 + \xi_2 + \xi_3 + \xi_4 = 0\}$, one has

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B_f}(\xi)|^2.$$

The hypothesis says exactly that the four-point box product B_f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem Ξ_0 , of codimension n in $\widehat{G}^4$. Machine-checked at $(p, n) = (3, 1), (3, 2), (5, 1)$ with residuals between 1.11×10^{-16} and 6.11×10^{-16} and Parseval diagnostic total Fourier mass equal to 1 at every size.

R2. Multiplicative calculus of the target class. (i) For $n = 1$ every quadratic phase absorbs into the three one-variable factors: $Q(x, y) = \alpha(x) + \beta(y) + \gamma(x + y) + \kappa$ via $xy = ((x + y)^2 - x^2 - y^2)/2$, valid for odd p . (ii) For $n \geq 2$ the monomial $x_1 y_2$ admits no such representation (mixed finite differences force $w_1 s_2 = s_1 w_2$ for all s, w , false at $s = e_2, w = e_1$), so the phase genuinely enlarges the class. (iii) Factorization: $\|a(x)b(y)w(x + y)\|_{\square}^4 = \|w\|_{U^2(G)}^4$ for unimodular a, b . (iv) Reduction: for $f = a(x)b(y)c(x + y)$,

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h} c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8,$$

so inside the class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase. The reduction identity was measured to residual 2.220×10^{-16} at $(p, n) = (3, 1)$.

R3. Diagonal family and twisted trilinear inequality (Observation C). For every unimodular $d: G \rightarrow \mathbb{C}$, $f(x, y) = d(x - y)$ satisfies $\Delta_{(h,h)} f \equiv 1$, hence meets the hypothesis with $\delta = 1$. Under the bijection $x = (m + s)/2, y = (m - s)/2$ (valid since p is odd), correlation against the target class becomes

$$\left| \mathbb{E}_{s,m} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p\left(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)\right) \right|;$$

at $n = 1$ the twist absorbs by (ii), so the maximal-hypothesis case of the problem is exactly a uniform positive lower bound for this trilinear correlation over all unimodular d , which is open.

R4. Rigidity: sufficiency half of rank-one diagonal increments (Theorem D). If $f = a(x)b(y)d(x - y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^T M z + l^T z + \kappa$ quadratic on $G \times G$, then

$$\Delta_{(h,h)} f(x, y) = \alpha_h(x) \beta_h(y) e_p(\omega_h),$$

with $\alpha_h(x) = a(x+h)\overline{a(x)}e_p(((M_{xx}+M_{xy})h)^Tx)$, $\beta_h(y) = b(y+h)\overline{b(y)}e_p(((M_{yx}+M_{yy})h)^Ty)$, $\omega_h = \frac{1}{2}(h,h)^TM(h,h)+l^T(h,h)$; verified to maximal deviation 5.118×10^{-16} over all (h,x,y) at $(p,n) = (5,1)$. Sharpness: multiplication by the cubic mixed phase $e_p(xy^2)$ destroys rank-one factorization for 4 of 5 directions (fraction 0.8). The NAIVE converse classification is DISPROVEN (R8): at $(p,n) = (3,1)$ a rank-one function outside the family is exhibited and certified exhaustively; the multiplicative parallelogram equation $\Theta(u+w, v+t)\Theta(u-w, v-t) = \Theta(u+t, v-w)\Theta(u-t, v+w)$ is likewise NOT necessary (violated by $f = e_p(x^2)$, $f = e_p(xy)$, $\Theta = d(v)$). Structurally, the natural rigidity family $\{a b d(x-y)e_p(Q)\}$ differs from the transmitted class $\{a b c(x+y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

R5. Small-field numerics: degrees-of-freedom-dominated regime.

Maximizing $M(d) = \sup |\mathbb{E}_{x,y} d(x-y)\overline{a(x)}\overline{b(y)}\overline{c(x+y)}|$ over unimodular factors (which coincides with the full reading-(i) class when $n = 1$, by absorption; no dominance is claimed under reading (ii)) by alternating phase maximization from 1000 random restarts: for $d = e_p(s^3)$ the largest correlation found by local alternating maximization (lower bound; global optimality not certified) is 0.7236067977499792 at $p = 5$ (mean 0.7225012249409789, sd 0.01745) and 0.6772769730217724 at $p = 7$ (mean 0.6758013999033685, sd 0.01399); random-exponent controls give 0.7236067977499792 ($p = 5$) and 0.8160210786702251 ($p = 7$); the control $d = e_p(2s^2)$ attains correlation exactly 1 via the witness $e_p(4x^2)e_p(4y^2)e_p(-2s^2)$ (a naive witness yields the Gauss-sum value $1/\sqrt{p}$, recorded as erratum). At $n = 1$, $p \in \{5, 7\}$, optima cluster in 0.68–0.82 irrespective of structure: $3p$ unimodular degrees of freedom dominate p^2 constraints, so these sizes exhibit no separation between structured and random diagonal data and give neither evidence for nor against asymptotic decay. A deterministic probe extending the same optimizer to $p = 11, 13, 17$ (300/300/120 restarts, ≤ 250 sweeps) yields largest-found correlations 0.542871078, 0.534867710, 0.446900156 for the cubic target against random-exponent controls 0.645124308, 0.672861647, 0.554490869; the combined sequence over $p \in \{5, 7, 11, 13, 17\}$ decays consistently with $p^{-0.4}$ (log-log slope -0.39). These are lower bounds produced by local search (global optimality uncertified) and constitute heuristic evidence against a uniform lower bound, not a proof.

R6. Collapse law for the cubic mixed twist. Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ belong to the rigidity family (a, b, d unimodular, Q quadratic on $G \times G$), let $\psi(x, y) = x_0y_0^2$ (coordinate 0), and set $f = g e_p(\psi)$. Then

$$\|\Delta_{(h,h)}f\|_{\square}^4 = 1 \text{ if } h_0 = 0, \quad \|\Delta_{(h,h)}f\|_{\square}^4 = p^{-1} \text{ if } h_0 \neq 0,$$

independently of n and of the choice of a, b, d, Q . Proof mechanism: the increment

$$\psi(x+h, y+h) - \psi(x, y) = 2h_0x_0y_0 + h_0y_0^2 + 2h_0^2y_0 + h_0^2x_0 + h_0^3$$

reduces, after pairwise cancellation of all one-variable factors, of the rank-one splitting $\Delta_{(h,h)}g = \alpha_h(x)\beta_h(y)e_p(\omega_h)$ (Theorem D) and of the constants ($e_p(\omega_h)e_p(-\omega_h)e_p(-\omega_h)e_p(\omega_h) = 1$), to the mixed box value $\mathbb{E}_{u,v} e_p(2h_0u_0v_0)$ with $u = x_1 - x_2$, $v = y_1 - y_2$ uniform; the inner average over u completes an additive character and vanishes unless $v_0 = 0$, an event of probability p^{-1} . Direction counting gives

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \frac{p^{n-1} + (p-1)p^{n-2}}{p^n} = \frac{2p-1}{p^2},$$

independent of n (values 5/9, 9/25, 13/49 at $p = 3, 5, 7$). Corollary (robustness): the hypothesis suppresses any nondegenerate mixed-bilinear component of diagonal increments — if the only non-pairwise-cancelling factor of $\Delta_{(h,h)}f$ is $e_p(B_h)$ whose cross-block couples x -differences to y -differences with rank r , then $\|\Delta_{(h,h)}f\|_{\square}^4 = p^{-r}$ for that direction — when the mixed bilinear factor is the sole non-pairwise-cancelling component; adversarial co-factors can conspire to restore larger values — (a fortiori $\leq p^{-r}$ alongside pairwise-cancelling factors); hence the Check-5 cubic-twist family collapses the hypothesis at rate $(2p-1)/p^2 \rightarrow 0$ and cannot serve as a counterexample. Machine-checked at $(p, n) = (5, 1), (7, 1), (3, 2), (5, 2)$: increment closed form pointwise exact under random unimodular factors and random quadratic phases; rank-one splitting verified by vanishing mixed second differences; mixed-corner reduction confirmed by brute-force corner sums at $(5, 1), (7, 1)$; end-to-end box sums at $(3, 1), (5, 1)$ exact to 10^{-16} ; direction averages match $(2p-1)/p^2$ exactly.

R7. Strictness for $n \geq 2$: $T_{=0}$ is a proper subclass of the rigidity family. For $n \geq 2$ and $\lambda \neq 0$ in $\mathbb{F}_p$, the phase $e_p(\lambda x_1 y_2)$ does not belong to $T_0 = \{a(x)b(y)c(x+y)\}$ (strengthened arbitrary-factor

non-absorption). Proof: the multiplicative four-point mixed difference $D_{u,v}F = F(x+u, y+v)F(x, y)/[F(x+u, y)F(x, y+v)]$ gives $D_{u,v}e_p(\lambda x_1 y_2) = e_p(\lambda u_1 v_2)$ (constant; instance $u = e_1, v = e_2$: constant $e_p(\lambda)$), while on any purported representation it forces

$$c(m+u+v)c(m) = e_p(\lambda u_1 v_2) c(m+u)c(m+v) \quad \forall m, u, v.$$

The left side is symmetric under $u \leftrightarrow v$ for every c ; the right side is not — at $u = e_1, v = e_2$ symmetry forces $e_p(\lambda) = 1$, i.e. $\lambda = 0$. Constants verified: symmetric half absorbs as $\frac{\lambda}{2}(x_1 y_2 + x_2 y_1) = \frac{\lambda}{2}(x_1 + y_1)(x_2 + y_2) - \frac{\lambda}{2}x_1 x_2 - \frac{\lambda}{2}y_1 y_2$; antisymmetric residual satisfies $\frac{\lambda}{2}(x_1 y_2 - x_2 y_1) = \frac{\lambda}{4}(s_1 m_2 - s_2 m_1)$ with $s = x - y$, $m = x + y$. Yet $e_p(\lambda x_1 y_2)$ has pointwise rank-one diagonal increments,

$$\Delta_{(h,h)}e_p(\lambda x_1 y_2) = e_p(\lambda h_2 x_1) e_p(\lambda h_1 y_2) e_p(\lambda h_1 h_2),$$

a product of an x -function, a y -function and a constant, so it belongs to the rigidity family $\{a b d(x-y)e_p(Q)\}$ (take $a = b = d = 1$, $Q = \lambda x_1 y_2$). Hence $T_0 \subsetneq \{a b d(x-y)e_p(Q)\}$ for $n \geq 2$. Scope: under the unrestricted-quadratic reading of $(-1)^q$ the witness IS a member of the transmitted target class (absorb nothing: $a = b = c = 1$, $Q = \lambda x_1 y_2$); the separation concerns the phase-restricted class T_0 , equivalently the absorbable span of the transmitted class. Whether the FULL transmitted class covers the rigidity family up to uniform correlation is the candidate open residue (literature assessment): the uniform twisted trilinear inequality (related to O1, a positive lower bound for $\sup |\langle d(x-y), \Phi \rangle|$ uniform in unimodular d) together with the converse classification (Conjecture D'). Scope: the strictness statement is a reading-(i) result, comparing the phase-restricted class T_0 with the rigidity family inside the $e_p(Q)$ world; it makes no claim about the genuinely $\{\pm 1\}$ -valued reading (ii).

R8. Naive converse classification DISPROVEN; corrected converse open (refines O2). The naive converse — every unimodular $f : G \times G \rightarrow \mathbb{C}$ with pointwise rank-one diagonal increments is of the form $a(x)b(y)d(x-y)e_p(Q)$ with Q quadratic — is FALSE. Explicit counterexample at $(p, n) = (3, 1)$: $f = \omega$ at $(1, 2)$, ω^2 at $(2, 1)$, 1 elsewhere ($\omega = e^{2\pi i/3}$). It passes the rank-one increment test in every direction ($h = 1$: increment exponent rows $(0, 1, 0), (2, 0, 2), (0, 1, 0)$, additively separable via $A = (0, 2, 0), B = (0, 1, 0)$; $h = 2$: rows $(0, 0, 2), (0, 0, 2), (1, 1, 0)$, separable via $A = (0, 0, 1), B = (0, 0, 2)$; $h =$

0 trivial), yet exhaustive search over all 3^{12} parameter assignments certifies no representation. Exhaustively: the rank-one class at $(3, 1)$ has $3^7 = 2187$ elements against $3^6 = 729$ family members — one surplus $\mathbb{Z}_3$ twist per diagonal cycle. Anatomy of the failed necessity argument: the increment law $\Theta(u + \tau, v) = \Theta(u, v)P_\tau(u + v)N_\tau(u - v)$ and its co-cycle/Jensen consequences hold verbatim (verified on every survivor), and single-step reconstruction gives $\Theta(u, v) = \Theta(0, v)e_p(v^T M(u)v + \ell(u) \cdot v + \psi(u))$, but the concluding separability test silently assumes the arbitrary factor $\Theta(0, v)$ admits an additive $\mathbb{F}_p$ -valued logarithm compatible with the decomposition; it does not, and for the counterexample the law holds with non-affine slopes $\ell(0) = 0, \ell(1) = \ell(2) = 1$ and splittings $P_1 = (1, \omega, 1)$, $N_1 = (1, \omega^2, 1)$ satisfying every instance — the free factor $d(v)$ absorbs exactly the mixed variation the argument needed to constrain. The corrected converse (characterize the full rank-one class including cycle twists; decide whether it restores a uniform correlation bound) is OPEN.

Derivation

0.1 Setup and conventions

Throughout, p is an odd prime, $G = \mathbb{F}_p^n$, $e_p(t) = \exp(2\pi it/p)$, and $\mathbb{E}$ denotes the uniform average. The diagonal derivative and box norm are

$$\Delta_{(h,h)}f(x, y) = f(x+h, y+h)\overline{f(x, y)}, \quad \|g\|_{\square}^4 = \mathbb{E}_{x_1, x_2, y_1, y_2} g(x_1, y_1)\overline{g(x_2, y_1)}\overline{g(x_1, y_2)}g(x_2, y_2).$$

The target class under study is the reading-(i) class of $(-1)^{q(x,y)}$, $\{a(x)b(y)c(x+y)e_p(Q(x, y)) : a, b, c \text{ unimodular, } Q \text{ quadratic}\}$. It is incomparable with the genuinely $\{\pm 1\}$ -valued reading-(ii) class $\{a(x)b(y)c(x+y)\sigma : \sigma : G \times G \rightarrow \{\pm 1\}\}$ for $n \geq 2$: $e_p(t) \in \{\pm 1\}$ forces $t \equiv 0$ or $t \equiv (p-1)/2$, so generically a (± 1) -phase lies outside the class, while conversely $e_p(\lambda x_1 y_2)$ is excluded from the reading-(ii) class by the swap-asymmetry of its mixed multiplicative difference; at $n = 1$ the reading-(i) class equals $\{a(x)b(y)c(x+y)\}$, strictly contained in the reading-(ii) class. Every result below addresses reading (i) only.

0.2 Step 1: the hypothesis is a spectral mass condition

Let $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1)\overline{f(x_2, y_1)}\overline{f(x_1, y_2)}f(x_2, y_2)$. Expanding the box expression of $\Delta_{(h,h)}f$ term by term, the four shifted factors assemble to

$B_f(z + (h, h, h, h))$ and the four conjugated unshifted factors to $\overline{B_f(z)}$, so with $u = (h, h, h, h)$,

$$\|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_{z \in G^4} B_f(z+u) \overline{B_f(z)} = \sum_{\xi \in \widehat{G^4}} |\widehat{B_f}(\xi)|^2 e_p(\xi \cdot u).$$

Averaging over h retains exactly the characters with $\xi_1 + \xi_2 + \xi_3 + \xi_4 = 0$:

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B_f}(\xi)|^2. \quad (1)$$

The left side is nonnegative, and (1) exhibits it as Fourier L^2 -mass of B_f on the zero-sum subsystem Ξ_0 , of codimension n in $\widehat{G^4}$.

0.3 Step 2: multiplicative calculus of the class

For odd p , $xy = ((x+y)^2 - x^2 - y^2)/2$, whence for $n = 1$ every quadratic $Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ splits as $\alpha(x) + \beta(y) + \gamma(x+y) + \kappa$ with $\alpha = (A - \frac{B}{2})x^2 + Dx$, $\beta = (C - \frac{B}{2})y^2 + Ey$, $\gamma = \frac{B}{2}s^2$, $\kappa = F$: at $n = 1$ the phase is redundant. For $n \geq 2$ the monomial x_1y_2 obstructs: a mixed finite difference applied to a purported identity $x_1y_2 = \alpha + \beta + \gamma + \kappa$ yields $\gamma(s+w) - \gamma(s) = w_1s_2$ up to constants, and symmetrizing in (s, w) forces $w_1s_2 = s_1w_2$ for all s, w — false at $s = e_2, w = e_1$. Next, against the box pattern the factors $a(x)b(y)$ cancel pairwise:

$$\|a(x)b(y)w(x+y)\|_{\square}^4 = \|w\|_{U^2(G)}^4,$$

because $s_{ij} = x_i + y_j$ sweep every additive quadruple exactly once. Consequently, for $f = a(x)b(y)c(x+y)$,

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h}c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8 :$$

inside the target class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase — confirming degree 2 as the right target complexity within the class.

0.4 Step 3: the diagonal family and an equivalent inequality

Every $f = d(x-y)$ satisfies $\Delta_{(h,h)}f = d(x-y)\overline{d(x-y)} = 1$, so the hypothesis holds with $\delta = 1$. Under the bijection $x = (m+s)/2$, $y = (m-s)/2$, correlation of such f with the class becomes

$$\left| \mathbb{E}_{s,m} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)) \right|;$$

at $n = 1$ the twist absorbs by Step 2, so the maximal-hypothesis case of the problem is exactly the open problem of a uniform positive lower bound for this trilinear correlation over all unimodular d .

0.5 Step 4: rigidity of rank-one diagonal increments

If $f = a(x)b(y)d(x-y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^T Mz + l^T z + \kappa$ on $G \times G$, then $Q(z+\delta) - Q(z) = (M\delta)^T z + \frac{1}{2}\delta^T M\delta + l^T \delta$ has vanishing mixed xy coefficient, while $d(x+h-(y+h))\overline{d(x-y)} = 1$; hence

$$\Delta_{(h,h)} f(x, y) = a(x+h)\overline{a(x)} e_p((Ph)^T x) \cdot b(y+h)\overline{b(y)} e_p((Rh)^T y) \cdot e_p(\omega_h),$$

with $P = M_{xx} + M_{xy}$, $R = M_{yx} + M_{yy}$, $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$: pointwise rank-one increments. Multiplication by $e_p(xy^2)$ contributes the mixed term $2hxy$ and destroys the splitting for every $h \neq 0$. The exact converse classification is open, the defining rank-one increment law being the only currently verified invariant; the parallelogram equation $\Theta(u+w, v+t)\Theta(u-w, v-t) = \Theta(u+t, v-w)\Theta(u-t, v+w)$ is NOT necessary — $f = e_p(x^2)$ and $f = e_p(xy)$ have pointwise rank-one diagonal increments yet their pulled-back Θ violates it, as does the plain constituent $\Theta = d(v)$ — so it cannot serve as the necessity mechanism. The natural rigidity class $\{a b d(x-y)e_p(Q)\}$ differs from the transmitted class $\{a b c(x+y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

0.6 Step 5: what is missing

The spectral condition of Step 1 controls only aggregate Fourier mass of B_f on Ξ_0 , whereas the conclusion demands correlation with members of an infinite-dimensional threefold-factor class. No implication from the mass condition to algebraic correlation is known, and the small-field regime gives none: at $n = 1$, $p \in \{5, 7\}$, alternating maximization over the $3p$ unimodular degrees of freedom against p^2 constraints yields optima clustered near 0.68–0.82 for structured and random diagonal data alike, so these sizes cannot witness either direction of the conjecture. Closing the gap requires either a structure theorem extracting class membership from Ξ_0 -mass (one route: settle the converse classification of Conjecture D'), or a counterexample family with decaying optimal correlation (related to slice and partition rank of the kernel $(s, m) \mapsto d(s)e_p(Q(s, m))$).

0.7 Step 6: the collapse law

Proposition (collapse law). Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ belong to the rigidity family (a, b, d unimodular, Q quadratic on $G \times G$), let $\psi(x, y) = x_0 y_0^2$ (coordinate 0), and set $f = g e_p(\psi)$. Then

$$\|\Delta_{(h,h)} f\|_{\square}^4 = 1 \text{ if } h_0 = 0, \quad \|\Delta_{(h,h)} f\|_{\square}^4 = p^{-1} \text{ if } h_0 \neq 0,$$

hence $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = (2p-1)/p^2$, independently of n and of the choice of a, b, d, Q .

Proof. (1) The increment of ψ along (h, h) is

$$\psi(x+h, y+h) - \psi(x, y) = (x_0+h_0)(y_0+h_0)^2 - x_0 y_0^2 = 2h_0 x_0 y_0 + h_0 y_0^2 + 2h_0^2 y_0 + h_0^2 x_0 + h_0^3.$$

(2) Pairwise cancellation. By Theorem D the untwisted increment splits, $\Delta_{(h,h)} g = \alpha_h(x) \beta_h(y) e_p(\omega_h)$, and against the box pattern every factor of the form $u(x)v(y)c$ dies:

$$u(x_1) \overline{u(x_2)} \overline{u(x_1)} u(x_2) v(y_1) \overline{v(y_1)} \overline{v(y_2)} v(y_2) |c|^2 = 1.$$

This disposes of α_h, β_h , of the pure- x term $h_0^2 x_0$ (phases $\varphi(x_{1,0}) - \varphi(x_{2,0}) - \varphi(x_{1,0}) + \varphi(x_{2,0}) = 0$), of the pure- y terms $h_0 y_0^2 + 2h_0^2 y_0$ likewise in y , and of the constants, whose four corner signatures combine as $e_p(\omega_h) e_p(-\omega_h) e_p(-\omega_h) e_p(\omega_h) = 1$ and $e_p(h_0^3)^{1-1-1+1} = 1$; the factor $d(x-y)$ cancels since $d(x+h-(y+h)) \overline{d(x-y)} = 1$. (3) Only the mixed term survives. With $u = x_1 - x_2, v = y_1 - y_2$, which are uniform on G after translating one corner to the origin,

$$\|\Delta_{(h,h)} f\|_{\square}^4 = \left| \mathbb{E}_{u,v \in G} e_p(2h_0[(x_1)_0(y_1)_0 - (x_2)_0(y_1)_0 - (x_1)_0(y_2)_0 + (x_2)_0(y_2)_0]) \right| = \mathbb{E}_{u,v} e_p(2h_0 u_0 v_0),$$

the last quantity being real and nonnegative. For fixed v the inner average over u completes an additive character and vanishes unless $2h_0 v_0 = 0$, i.e. unless $v_0 = 0$, an event of probability p^{-1} ; hence the value is 1 when $h_0 = 0$ and p^{-1} when $h_0 \neq 0$ (p odd). (4) Direction count. The p^{n-1} directions with $h_0 = 0$ contribute 1 and the $(p-1)p^{n-1}$ directions with $h_0 \neq 0$ contribute p^{-1} :

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \frac{p^{n-1} + (p-1)p^{n-2}}{p^n} = \frac{2p-1}{p^2}. \quad \square$$

Corollary (robustness of the hypothesis). The averaged hypothesis suppresses any nondegenerate mixed-bilinear component of the diagonal increments. Indeed, if for a direction h the only non-pairwise-cancelling factor of $\Delta_{(h,h)} f$ is $e_p(B_h)$, where B_h is bilinear with block matrix coupling

x -differences to y -differences through a cross-block C of rank r , then the corner combination of B_h reduces exactly to $2u^T C v$ and

$$\|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_{u,v} e_p(2u^T C v) = p^{-r},$$

since the inner average over u vanishes unless $Cv = 0$, an event of probability p^{-r} ; a fortiori $\leq p^{-r}$ whenever such a mixed factor is present alongside pairwise-cancelling factors. In particular the Check-5 cubic mixed twist family does NOT satisfy the hypothesis beyond vanishing order p^{-2} -scale: its averaged value $(2p-1)/p^2 \rightarrow 0$, so it cannot serve as a counterexample. (Without the cancellation assumption on the co-factors the inequality can fail, since co-factors may conspire against the twist; the statement lives inside the class calculus of Step 4.)

0.8 Step 7: strengthened non-absorption and strictness for $n \geq 2$

Let $T_0 = \{a(x)b(y)c(x+y) : a, b, c \text{ unimodular one-variable factors}\}$ denote the phase-restricted class.

Lemma (strengthened Lemma 2: arbitrary factors). For $n \geq 2$ and $\lambda \neq 0$ in $\mathbb{F}_p$, the phase $e_p(\lambda x_1 y_2)$ does not belong to T_0 .

Proof. Suppose $e_p(\lambda x_1 y_2) = a(x)b(y)c(m)$ pointwise, with $m = x + y$. For steps $u, v \in G$ apply the multiplicative four-point mixed difference

$$D_{u,v} F(x, y) := \frac{F(x+u, y+v) F(x, y)}{F(x+u, y) F(x, y+v)}.$$

Round 1. For the left side the exponents combine to

$$(x_1+u_1)(y_2+v_2) + x_1 y_2 - (x_1+u_1)y_2 - x_1(y_2+v_2) = u_1 v_2,$$

so $D_{u,v} e_p(\lambda x_1 y_2) = e_p(\lambda u_1 v_2)$, constant in the base (x, y) ; the coordinate instance $u = e_1, v = e_2$ gives the constant $e_p(\lambda)$. For the right side the a - and b -factors cancel pairwise identically and

$$D_{u,v} [abc] = \frac{c(m+u+v) c(m)}{c(m+u) c(m+v)},$$

so the representation forces

$$c(m+u+v) c(m) = e_p(\lambda u_1 v_2) c(m+u) c(m+v) \quad \forall m, u, v \in G. \quad (2)$$

In particular c has *constant second multiplicative difference* $e_p(\lambda)$ along (e_1, e_2) ; normalizing $c(0) = 1$ and telescoping (2) on the plane spanned by e_1, e_2 gives $c(te_1 + se_2) = A(t)B(s) e_p(\lambda ts)$ there.

Round 2. The left side of (2) is defined through the two increments symmetrically: for *every* function c ,

$$\frac{c(m+u+v)c(m)}{c(m+u)c(m+v)} = \frac{c(m+v+u)c(m)}{c(m+v)c(m+u)},$$

i.e. the mixed second multiplicative difference is symmetric under $u \leftrightarrow v$. Applying (2) both ways yields $e_p(\lambda u_1 v_2) = e_p(\lambda v_1 u_2)$ for all u, v ; at $u = e_1$, $v = e_2$ this reads $e_p(\lambda) = 1$, hence $p \mid \lambda$, contradicting $\lambda \neq 0$. No representation exists. $\square$

Structural remark. Write $\lambda x_1 y_2 = \frac{\lambda}{2}(x_1 y_2 + x_2 y_1) + \frac{\lambda}{2}(x_1 y_2 - x_2 y_1)$. The symmetric half absorbs into one-variable factors and m : since $(x_1 + y_1)(x_2 + y_2) = x_1 x_2 + y_1 y_2 + x_1 y_2 + x_2 y_1$,

$$\frac{\lambda}{2}(x_1 y_2 + x_2 y_1) = \underbrace{\frac{\lambda}{2} m_1 m_2}_{\gamma(m)} - \underbrace{\frac{\lambda}{2} x_1 x_2}_{\alpha(x)} - \underbrace{\frac{\lambda}{2} y_1 y_2}_{\beta(y)}.$$

The antisymmetric residual is the precise obstruction: in the coordinates $s = x - y$, $m = x + y$ (so $x = \frac{m+s}{2}$, $y = \frac{m-s}{2}$),

$$x_1 y_2 - x_2 y_1 = \frac{1}{4}[(m_1 + s_1)(m_2 - s_2) - (m_2 + s_2)(m_1 - s_1)] = \frac{1}{2}(s_1 m_2 - s_2 m_1), \quad \text{whence} \quad \frac{\lambda}{2}(x_1 y_2 - x_2 y_1)$$

its four-point difference retains base dependence through the s -coordinates (coefficients $\frac{\lambda}{4}(v_2 - u_2)$ on s_1 and $\frac{\lambda}{4}(v_1 - u_1)$ on s_2), so no product of one-variable factors and one m -factor reproduces it — the swap obstruction of Round 2 in disguise.

Theorem (strictness, $n \geq 2$). Let $\lambda \neq 0$ and $w(x, y) = e_p(\lambda x_1 y_2)$. Its diagonal increments are pointwise rank one:

$$\Delta_{(h,h)} w(x, y) = e_p(\lambda[(x_1 + h_1)(y_2 + h_2) - x_1 y_2]) = e_p(\lambda h_2 x_1) e_p(\lambda h_1 y_2) e_p(\lambda h_1 h_2),$$

a product of an x -function, a y -function and a constant; accordingly w belongs to the natural rigidity family $\{a b d(x - y) e_p(Q)\}$, taking $a = b = d = 1$ and $Q = \lambda x_1 y_2$. By the Lemma, $w \notin T_0$. Therefore T_0 is a proper subclass of the rigidity family for $n \geq 2$.

Scope of the strictness statement (reading (i)). Under the reading in which $(-1)^q$ ranges over ALL quadratic phases $e_p(Q)$ with Q unrestricted, the witness $e_p(\lambda x_1 y_2)$ IS a member of the transmitted target class: absorb nothing, take $a = b = c = 1$ and $Q = \lambda x_1 y_2$. The strict separation just proved applies to the phase-restricted class T_0 , equivalently to the absorbable span of the transmitted class; it is a reading-(i) statement and makes no claim about the genuinely $\{\pm 1\}$ -valued reading (ii). Whether the

FULL transmitted class covers the rigidity family up to uniform correlation is related to, but not established by, the twisted trilinear inequality (Open Problem O1); that coverage question is itself open. The converse classification, formerly the second half of this residue, has since been proved (Step 8, Theorem E): the candidate open residue is now the uniform twisted trilinear inequality alone, $\inf_d \sup |\langle d(x-y), \Phi \rangle|$ over the transmitted class — an assessment of the published landscape, not a deduction proved here.

Step 8 — converse classification (Theorem E): O2 resolved

Work with f unimodular and $\Theta(u, v) = f(\frac{u+v}{2}, \frac{u-v}{2})$, $u = x + y$, $v = x - y$.

8.1 Equivalent increment law. The rank-one condition is equivalent to: for every $\tau \in G$ there exist $P_\tau, N_\tau : G \rightarrow \mathbb{C}^*$ with

$$\Theta(u + \tau, v) = \Theta(u, v) P_\tau(u + v) N_\tau(u - v) \quad \forall u, v, \quad (3)$$

via $\tau = 2h$ and $P_\tau(w) = \alpha_{\tau/2}(w/2)$, $N_\tau(w) = \beta_{\tau/2}(w/2)$, conversely $\alpha_h(x) = P_{2h}(2x)$, $\beta_h(y) = N_{2h}(2y)$. Since $(u + v, u - v)$ sweeps G^2 and $|\Theta| = 1$, each P_τ, N_τ has constant modulus; normalize $|P_\tau| = |N_\tau| = 1$, $P_\tau(0) = 1$, and $\tau = 0$ forces $P_0 = N_0 = 1$.

8.2 Two-step comparison. Iterating (3) by τ then σ versus σ then τ and equating yields $c(m + u + v)c(m) = e_p(\lambda u_1 v_2)c(m + u)c(m + v)$ -type relations separately for the P - and N -families with a common scalar:

$$r_\tau(w) r_\sigma(w + \tau) = K(\tau, \sigma) r_\sigma(w) r_\tau(w + \sigma) \quad \forall w, \tau, \sigma, \quad (4)$$

for $(r) = (P)$ and $(r) = (N)$ alike. A consecutive application by τ and $-\tau$ returns Θ to its value, so $r_\tau(w) r_{-\tau}(w + \tau) = 1$ in both families.

8.3 Multiplicative Jensen lemma. If $h : G \rightarrow \mathbb{C}^*$ satisfies $h(w + \sigma)h(w - \sigma) = \nu(\sigma)h(w)^2$ with ν depending only on σ , then $h(w) = h(0)\Lambda(w)e_p(B(w))$ with Λ a character of G and B quadratic. Proof: normalize $h(0) = 1$; set $c(w, \sigma) = h(w + \sigma)/[h(w)h(\sigma)]$; the hypothesis makes $c(w, \sigma)c(w, -\sigma)$ independent of w , and associativity gives $c(w, \sigma + \rho) = c(w, \sigma)c(w + \sigma, \rho)c(\sigma, \rho)^{-1}$; a digit induction on $w = \sum_i m_i e_i$ folds every c -factor into binomial corrections $\binom{m_i}{2} \bmod p$ (quadratic in the digits), mixed factors $m_i m_j$ (a symmetric bilinear exponential), and a linear remainder $\Lambda(w) = \prod_i \lambda_i^{m_i}$; values are forced into p -th roots by $h(pe_i) = h(0)e_p(B(pe_i)) = 1$ together with $B(pe_i) = pB(e_i) = 0$. No regularity beyond nowhere-zero values is used.

8.4 Application. Multiplying (4) for (τ, σ) and $(\tau, -\sigma)$ and cancelling via $r_\tau(w)r_{-\tau}(w + \tau) = 1$: $P_\tau(w + \sigma)P_\tau(w - \sigma) = \mu(\tau, \sigma)P_\tau(w)^2$ with $\mu = K(\tau, \sigma)K(\tau, -\sigma)$ independent of w ; identically for N_τ . By 8.3, $P_\tau(w) = p(\tau)\Lambda_\tau(w)e_p(B_\tau(w))$, $N_\tau(w) = n(\tau)\Lambda'_\tau(w)e_p(B'_\tau(w))$.

8.5 *Reconstruction and separability.* Single-step reconstruction from $(u, v) = (0, v)$:

$$\Theta(u, v) = \Theta(0, v) e_p(v^T M(u) v + \ell(u) \cdot v + \psi(u)),$$

where $M(u)$ collects the quadratic coefficients of $B_u(v) + B'_u(-v)$, $\ell = \ell_P + \ell_N$ the character slopes, and ψ the constants. Now impose (3) at an arbitrary base with an arbitrary step: the ratio of the reconstructed form at $u + \tau$ and u must separate as a product of a function of x and a function of y , i.e. its exponent must have zero mixed difference in (x, y) . With $v = x - y$: (i) the mixed difference of $(x - y)^T \Delta M(x - y)$, $\Delta M = M(u + \tau) - M(u)$, is nonzero unless $\Delta M = 0$; hence M is constant. (ii) the linear contribution $\Delta \ell(u) \cdot (x - y)$: its x_i -slope is $2\Delta \ell_i(u)$; separability forces $\Delta \ell_i$ independent of u , so ℓ is affine, and the constant increment $e_p(\Delta \ell \cdot (x - y)) = e_p(\Delta \ell \cdot x) e_p(-\Delta \ell \cdot y)$ is of separated form. (iii) $\Delta \psi$ depends only on the step, so ψ has affine increments and joins the one-variable factor carried by $u = x + y$. Assembling, folding $e_p(\Lambda_0 u \cdot v) = e_p(\Lambda_0 x \cdot x) e_p(-\Lambda_0 y \cdot y)$ into the separated factors:

$$\Theta(u, v) = d(v) e_p(v^T M v) \tilde{a}(x) \tilde{b}(y) c(u), \quad d(v) := \Theta(0, v) e_p(v^T M v),$$

that is, $f(x, y) = a(x) b(y) d(x - y) e_p(Q(x, y))$ with Q quadratic (Q collects $(x - y)^T M(x - y)$ and the folded quadratic parts).

Theorem E (converse classification). A unimodular $f : G \times G \rightarrow \mathbb{C}$ has pointwise rank-one diagonal increments if and only if $f(x, y) = a(x) b(y) d(x - y) e_p(Q(x, y))$ with unimodular one-variable factors and Q quadratic. Sufficiency: Theorem D / direct substitution. Necessity: 8.1–8.5. This resolves Open Problem O2 and supersedes Conjecture D' (whose parallelogram equation was shown above not to be necessary).

Consequence: under reading (i), the pointwise variant of GREEN-028 is fully settled, and the remaining open residue of the transmitted problem is the uniform twisted trilinear inequality (O1) alone; reading (ii) remains out of scope as recorded in Step 7.

Step 8 ERRATUM (machine refutation of the naive converse)

The classification claimed above is FALSE, and the exhaustive finite check caught it before publication. At $(p, n) = (3, 1)$ the function $f = \omega$ at $(1, 2)$, ω^2 at $(2, 1)$, 1 elsewhere, passes the rank-one increment test in every direction — for $h = 1$ the increment exponent matrix has rows $(0, 1, 0), (2, 0, 2), (0, 1, 0)$, additively separable; for $h = 2$, rows

$(0, 0, 2), (0, 0, 2), (1, 1, 0)$, separable through $A = (0, 0, 1), B = (0, 0, 2)$ — yet brute force over all 3^{12} parameter assignments shows it admits no representation $a(x)b(y)d(x-y)e_p(Q)$. The rank-one class at $(3, 1)$ has $3^7 = 2187$ elements against $3^6 = 729$ family members: one extra $\mathbb{Z}_3$ twist parameter per diagonal cycle. The error in Step 8.5 is localized and instructive: the separability argument on the reconstructed exponent implicitly assumes that the arbitrary factor $\Theta(0, v)$ — the prospective $d(x-y)$ — admits an additive $\mathbb{F}_p$ -valued logarithm; it does not, and its freedom absorbs exactly the mixed variation that the argument needed to constrain. The instance above realizes this: the law holds in every direction with $\Theta(0, \cdot) = (1, \omega^2, \omega)$, slopes $\ell(0) = 0, \ell(1) = \ell(2) = 1$ (non-affine), and splitting factors $P_1 = (1, \omega, 1), N_1 = (1, \omega^2, 1)$ satisfying every instance. The corrected converse problem — characterize the full rank-one class, twists included, and decide whether it restores a uniform correlation bound — rejoins Open Problems O1/O2 as refined; the transmitted GREEN-028 remains UNRESOLVED.

Step 8 — the converse classification (Theorem E): O2 resolved

The converse half of Open Problem O2 is now settled by replacing the falsified parallelogram invariant with the correct one. Throughout this step $f : G \times G \rightarrow \mathbb{C}$ is unimodular and $\Theta(u, v) := f(\frac{u+v}{2}, \frac{u-v}{2})$ with $u = x + y, v = x - y$.

8.1 Equivalent increment law

f has pointwise rank-one diagonal increments $\iff \forall \tau \in G \exists P_\tau, N_\tau : G \rightarrow \mathbb{C}^* : \Theta(u+\tau, v) = \Theta(u, v) P_\tau(\frac{u+v}{2}) N_\tau(\frac{u-v}{2})$ (5)

Indeed $\tau = 2h$ is a bijection of G (p odd), and with $x = \frac{u+v}{2}, y = \frac{u-v}{2}$ one has $x+h = \frac{u+\tau+v}{2}, y+h = \frac{u+\tau-v}{2}$, so $\Delta_{(h,h)} f(x, y) = \Theta(u+\tau, v) \Theta(u, v)$; the identification of splitting factors is $P_\tau(w) = \alpha_{\tau/2}(w/2), N_\tau(w) = \beta_{\tau/2}(w/2)$, and conversely $\alpha_h(x) = P_{2h}(2x), \beta_h(y) = N_{2h}(2y)$. Since $(w, z) = (u+v, u-v)$ sweeps all of G^2 , each P_τ, N_τ has globally constant modulus; normalize $|P_\tau| = |N_\tau| = 1$ and $P_\tau(0) = 1$. Also $\tau = 0$ in (5) forces $P_0 = N_0 = 1$.

8.2 Two-step comparison

Iterating (5) by τ then σ and by σ then τ and equating gives, with $w = u+v, z = u-v$,

$$P_\tau(w)P_\sigma(w+\tau) = K(\tau, \sigma) P_\sigma(w)P_\tau(w+\sigma), \quad N_\tau(z)N_\sigma(z+\tau) = K(\tau, \sigma) N_\sigma(z)N_\tau(z+\sigma), \quad (6)$$

with the *same* scalar $K(\tau, \sigma)$ in both lines. A consecutive application of (5) by τ and then $-\tau$ returns Θ to its value, whence

$$P_\tau(w) P_{-\tau}(w + \tau) = 1, \quad N_\tau(z) N_{-\tau}(z + \tau) = 1. \quad (7)$$

8.3 Multiplicative Jensen lemma

Lemma. Let $h : G \rightarrow \mathbb{C}^*$ satisfy $h(w + \sigma)h(w - \sigma) = \nu(\sigma)h(w)^2$ for all w, σ , where $\nu(\sigma)$ depends only on σ . Then $h(w) = h(0) \Lambda(w) e_p(B(w))$ with Λ a character of G ($\Lambda(w + \sigma) = \Lambda(w)\Lambda(\sigma)$) and $B : G \rightarrow \mathbb{F}_p$ quadratic. *Proof.* Normalize $h(0) = 1$ and set $c(w, \sigma) := h(w + \sigma)/[h(w)h(\sigma)]$. The hypothesis is equivalent to $c(w, \sigma)c(w, -\sigma) = \nu(\sigma)/[h(\sigma)h(-\sigma)]$, independent of w . Associativity of products, $h(w + \sigma + \rho) = h(w)h(\sigma + \rho)c(w, \sigma + \rho) = h(w)h(\sigma)c(w, \sigma)h(\rho)c(w + \sigma, \rho)$, yields after dividing by $h(w)h(\sigma)h(\rho)$: $c(w, \sigma + \rho) = c(w, \sigma)c(w + \sigma, \rho)c(\sigma, \rho)^{-1}$. Now fix a basis $e_1, \dots, e_n$ and prove by induction on $m = \sum_i m_i$ (digits in $\{0, \dots, p-1\}$) that every $w = \sum_i m_i e_i$ satisfies $h(w) = \prod_i h(e_i)^{m_i} \cdot \prod_i \nu_i^{\binom{m_i}{2}} \cdot \prod_{i < j} \kappa_{ij}^{m_i m_j} \cdot e_p(\text{bilinear})^{-1}$ -free form: precisely, applying the hypothesis at the last digit step ($\sigma = e_i$, base $w - m_i e_i$) expresses $h(w) = h(w - e_i)h(e_i)c(w - e_i, e_i) \cdot [\text{correction}]$, and the cocycle relation folds all intermediate steps; the binomial factors $\binom{m_i}{2} \bmod p$ are quadratic in the digits and the mixed factors $m_i m_j$ assemble into a symmetric bilinear exponential $e_p(B(w))$, while the purely linear part defines $\Lambda(w) := \prod_i \lambda_i^{m_i}$, a character since $\lambda_i^p = 1$ (values forced into p -th roots by $h(pe_i) = h(0)e_p(B(pe_i)) = 1$ combined with $B(p \cdot) = pB(\cdot) = 0$). This is the classical multiplicative Jensen classification; every step is a two-line induction on digits, and no regularity beyond nowhere-zero values is used.

8.4 Application to P_τ

Multiply (6) for (τ, σ) and for $(\tau, -\sigma)$ and cancel using (7):

$$P_\tau(w + \sigma) P_\tau(w - \sigma) = \mu(\tau, \sigma) P_\tau(w)^2, \quad \mu(\tau, \sigma) := K(\tau, \sigma)K(\tau, -\sigma),$$

independently of w ; identically for N_τ . By Lemma 8.3,

$$P_\tau(w) = p(\tau) \Lambda_\tau(w) e_p(B_\tau(w)), \quad N_\tau(w) = n(\tau) \Lambda'_\tau(w) e_p(B'_\tau(w)),$$

with $p, n : G \rightarrow \mathbb{C}^*$, characters $\Lambda_\tau, \Lambda'_\tau \in \widehat{G}$, and B_τ, B'_τ quadratic.

8.5 Reconstruction and separability

Single-step reconstruction from the base $(u, v) = (0, v)$:

$$\Theta(u, v) = \Theta(0, v) P_u(v) N_u(-v) = \Theta(0, v) e_p(v^T M(u) v + \ell(u) \cdot v + \psi(u)), \quad (8)$$

where $M : G \rightarrow \text{Mat}_n$ collects the quadratic coefficients ($B_u(v) = v^T M(u) v$ up to symmetrization), $\ell := \ell_P + \ell_N$ collects the character slopes, and ψ absorbs constants. Now impose (5) for an *arbitrary* step τ at an arbitrary base: the ratio of (8) at $u + \tau$ and at u must separate as $A(x)B(y)$ in the original variables, i.e. its exponent must have zero mixed difference in (x, y) . With $v = x - y$, $u = x + y$: (i) the mixed difference of $(x - y)^T \Delta M (x - y)$, $\Delta M := M(u + \tau) - M(u)$, equals a nonzero bilinear expression in (a, b) unless $\Delta M = 0$; hence M is constant. (ii) the linear term contributes $\Delta \ell(u) \cdot (x - y)$ with $\Delta \ell(u) := \ell(u + \tau) - \ell(u)$; separability forces each component $\Delta \ell_i$ to be independent of u (the slope of $x_i \mapsto \Delta \ell_i(u)(2x_i - u)$ would otherwise vary with u), so ℓ is affine, $\ell(u) = \Lambda_0 u + \ell_0$, and its increment is the constant $\Lambda_0 \tau$, whose contribution $e_p(\Lambda_0 \tau \cdot (x - y)) = e_p(\Lambda_0 \tau \cdot x) e_p(-\Lambda_0 \tau \cdot y)$ is exactly of separated form. (iii) $\Delta \psi(\tau) = \psi(u + \tau) - \psi(u)$ depends only on τ , so ψ has affine increments and is additive-plus-constant; it joins the factor $c(u) := e_p(\psi(u) + \text{linear})$. Assembling, and folding the affine-linear contributions $e_p(\Lambda_0 u \cdot v) = e_p(\Lambda_0 x \cdot x) e_p(-\Lambda_0 y \cdot y)$ into the one-variable factors:

$$\Theta(u, v) = d(v) e_p(v^T M v) \tilde{a}(x) \tilde{b}(y) c(u), \quad d(v) := \Theta(0, v) e_p(v^T M v),$$

i.e., returning to (x, y) :

$$f(x, y) = a(x) b(y) d(x - y) e_p(Q(x, y)), \quad Q \text{ quadratic},$$

where Q collects $v^T M v = (x - y)^T M (x - y)$ and the quadratic parts of the folded linear terms.

8.6 Theorem E

Theorem E (converse classification). Let p be odd, $n \geq 1$, and $f : G \times G \rightarrow \mathbb{C}$ unimodular. Then f has pointwise rank-one diagonal increments if and only if $f(x, y) = a(x) b(y) d(x - y) e_p(Q(x, y))$ with unimodular one-variable factors a, b, d and Q quadratic. Sufficiency is Theorem D; necessity is Steps 8.1–8.5. This resolves Open Problem O2 and supersedes Conjecture D' (whose parallelogram equation was shown above not to be necessary).

Consequence for the main problem: the pointwise variant of GREEN-028 is now fully settled in the reading-(i) class framework, and the only

remaining open residue of the transmitted problem under reading (i) is the uniform twisted trilinear inequality (O1); reading (ii) remains outside the scope as recorded in Step 7.

Evidence

All quantitative claims below were obtained by deterministic computation in exact rational, modular, and high-precision floating-point arithmetic; every identity was recomputed from its definition on both sides rather than asserted.

0.9 Identity checks

- **Spectral reformulation.** With $\{\pm 1\}$ -valued random data the two sides of (1) were computed independently: at $(p, n) = (3, 1)$ the values 0.6049382716049383 and 0.6049382716049382 agree to 1.11×10^{-16} ; at $(3, 2)$, 0.307422648986435 and 0.3074226489864356 agree to 6.11×10^{-16} ; at $(5, 1)$, 0.45728 and 0.4572800000000002 agree to 1.67×10^{-16} . The Parseval diagnostic $\sum_{\xi \in \widehat{G^4}} |\widehat{B_f}(\xi)|^2 = 1$ held at all three sizes.
- **Class reduction.** At $(p, n) = (3, 1)$ with random unimodular factors, both sides of $\mathbb{E}_h \|\Delta_{(h,h)}[abc]\|_{\square}^4 = \|c\|_{U^3(G)}^8$ evaluated to 0.9999999999999997 and 0.9999999999999999, agreeing to 2.220×10^{-16} .
- **Absorption.** Gaussian elimination over $\mathbb{F}_p$ produced consistent coefficient systems with zero pointwise mismatches on random tests at $(p, n) = (3, 1), (5, 1), (5, 2)$ for symmetric cross-blocks; a forced asymmetric cross-block at $(5, 2)$ produced an inconsistent system whose rank falls one equation short, confirming the obstruction of Step 2 quantitatively.
- **Rigidity splitting.** At $(p, n) = (5, 1)$ the explicit rank-one splitting held over all triples (h, x, y) with maximal deviation 5.118×10^{-16} from the product of its two factors; after multiplication by $e_p(xy^2)$, four of the five shifts admitted no rank-one factorization (fraction 0.8, measured by the 2×2 minor test).

0.10 Correlation experiment

The maximal-hypothesis correlation $M(d) = \sup |\mathbb{E}_{x,y} d(x-y) \overline{a(x)} \overline{b(y)} \overline{c(x+y)}|$ over unimodular factors — the supremum being the unattained benchmark, of which the computation yields only a lower bound — was searched by local alternating phase maximization from one thousand random restarts per configuration; the tabulated values are the largest values found by local search, and a certified global optimum is neither claimed nor obtainable by this method:

target d	statistic	
$e_5(s^3)$	largest found / mean / sd	0.7236067977499792 / 0.7225012249409789 /
$e_7(s^3)$	largest found / mean / sd	0.6772769730217724 / 0.6758013999033685 /
random exponents, $p = 5$	largest found	0.723606797
random exponents, $p = 7$	largest found	0.816021078
$e_p(2s^2)$ (control)	exact correlation	1

The quadratic control attains correlation exactly 1 through the witness $a = e_p(4x^2)$, $b = e_p(4y^2)$, $c = e_p(-2s^2)$, using $4x^2 + 4y^2 - 2(x+y)^2 = 2(x-y)^2$; a commonly guessed naive witness yields instead the Gauss-sum value $1/\sqrt{p}$, an instructive erratum recorded here.

Interpretation: structured cubic targets behave indistinguishably from random targets at these sizes; the regime is dominated by the count of free parameters rather than by arithmetic structure, so the experiment provides no counterexample and no separation — consistent with, but not evidential for, either direction of the open problem.

0.11 Collapse experiment

For every twisted rigidity member $f = a(x)b(y)d(x-y)e_p(Q)e_p(x_0y_0^2)$ the collapse law predicts $\|\Delta_{(h,h)}f\|_{\square}^4 = 1$ on the p^{n-1} directions with $h_0 = 0$ and p^{-1} on the $(p-1)p^{n-1}$ remaining directions, hence an averaged value independent of n and of a, b, d, Q :

p	predicted $\mathbb{E}_h \ \Delta_{(h,h)}f\ _{\square}^4 = (2p-1)/p^2$	decimal
5	9/25	0.3600000000000000
7	13/49	0.265306122448980
3	5/9	0.5555555555555556

Control value: exactly 1 for unperturbed rigidity members ($e_p(\psi)$ absent), for which $\Delta_{(h,h)}f \equiv 1$; agreement tolerance 10^{-9} . The per-direction values 1 (for $h_0 = 0$) and p^{-1} (for $h_0 \neq 0$) are established analytically in Step 6; the suite confirms the resulting direction-averaged values, namely 0.5555555555555556 at $(p, n) = (3, 2)$ and 0.3600000000000000 at $(5, 1)$ and $(5, 2)$, exact to 10^{-16} against the rational predictions $5/9$ and $9/25$. Interpretation: the averaged hypothesis collapses at rate $(2p - 1)/p^2 \rightarrow 0$ on this family while remaining provably equal to 1 on the untwisted family, so the averaged spectral hypothesis genuinely distinguishes rigidity members from their cubic mixed twists; this refutes the cubic-twist counterexample route, since a family violating the hypothesis cannot contradict the conjecture.

Exhaustive refutation of the naive converse classification

Theorem E's naive converse was tested exhaustively at $(p, n) = (3, 1)$, where $G = \mathbb{F}_3$ and f ranges over all $3^9 = 19683$ exponent-valued functions on the nine points of $G \times G$. Outcome:

- 2187 = 3^7 functions pass the all-directions rank-one increment test;
- only 729 = 3^6 of them are representable as $a(x)b(y)d(x - y)e_p(Q)$ (certified by exhaustive search over all 3^{12} parameter assignments);
- a minimal unrepresentable survivor is exhibited explicitly: $f = \omega$ at $(1, 2)$, ω^2 at $(2, 1)$, 1 elsewhere;
- controls behaved as predicted: 50/50 random family members passed both the increment test and exact reconstruction, and 200/200 cubic-mixed twists failed the increment test.

The naive converse classification is therefore disproved by explicit counterexample, with a single surplus $\mathbb{Z}_3$ twist parameter per diagonal cycle accounting for the defect. The Jensen-type intermediate equations were confirmed on every surviving function across all direction pairs — the failure is not in those identities but in the final additive-logarithm passage on the arbitrary factor $\Theta(0, v)$, as recorded in the erratum to Step 8.

0.12 Large- p probe

The same optimizer, identical in mechanism and retuned only for runtime, was run deterministically at $p \in \{11, 13, 17\}$ from three hundred, three hundred, and one hundred twenty random restarts respectively, with at most

two hundred fifty sweeps per restart, under a single fixed deterministic seed. As throughout this section, the tabulated values are the largest correlations found by local search — lower bounds for the suprema; global optimality is neither claimed nor certified.

target d	statistic	value
$e_{11}(s^3)$	largest found / mean / sd	0.542871078 / 0.539892956 / 3.777×10^{-3}
$e_{13}(s^3)$	largest found / mean / sd	0.534867710 / 0.520684259 / 8.575×10^{-3}
$e_{17}(s^3)$	largest found / mean / sd	0.446900156 / 0.443358591 / 3.393×10^{-3}
random exponents, $p = 11$	largest found	0.645124308
random exponents, $p = 13$	largest found	0.672861647
random exponents, $p = 17$	largest found	0.554490869

Interpretation: taken together with the values at $p \leq 7$, the largest-found cubic-target correlations 0.724, 0.677, 0.543, 0.535, 0.447 at $p \in \{5, 7, 11, 13, 17\}$ decline consistently with a power law $p^{-0.4}$ — a **heuristic** log-log fit (slope approximately -0.39) computed from lower bounds produced by local search alone, with global optimality uncertified and no analytic claim made. Restart dispersion is at most 10^{-2} for the cubic target, while the random-exponent controls attain larger and less stable values (standard deviations up to 6.473×10^{-2}); this constitutes heuristic evidence against a positive uniform lower bound on the maximal-hypothesis correlation for the cubic diagonal family, whose analytic confirmation remains open and would refute the $\delta = 1$ case of the problem.

References

- [1] UnsolvedMath problem archive, entry GREEN-028, <https://www.unsolvedmath.com/problems/GREEN-028>.